

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 16: Control of Aircraft & Emergencies

Syllabus Source: CAE Oxford (pp. 331-356) | Pass Target: >= 90% AviationExam

1. SPEED CONTROL LIMITATIONS (ATC)

ATC may apply speed control to maintain separation. However, speed adjustments must not be applied or continued within 4 NM of the runway threshold on final approach. Speed reductions should not normally be below 160 kt IAS for jet aircraft on intermediate/final approach.

2. FUEL EMERGENCIES: MINIMUM FUEL VS MAYDAY FUEL

[!] EXAM TRAP (AVIATIONEXAM) - MINIMUM FUEL vs MAYDAY MAYDAY MAYDAY FUEL

- 'MINIMUM FUEL': Informs ATC that all planned aerodrome options are committed and that any change to clearance may result in landing with less than planned final reserve fuel. IT IS NOT AN EMERGENCY and DOES NOT CONFER PRIORITY. - 'MAYDAY MAYDAY MAYDAY FUEL': Distress message. Declares that calculated usable fuel on landing will be LESS than the mandatory final reserve fuel. Confers ABSOLUTE PRIORITY.

3. RADIO COMMUNICATIONS FAILURE PROCEDURES (SERA.8035)

- Transponder Setting: Squawk Mode A Code 7600 immediately.
- In VMC: Continue to fly in VMC; land at the nearest suitable aerodrome; report arrival to the appropriate ATS unit by the most expeditious means.
- In IMC (Controlled Airspace): Maintain last assigned speed and level for 7 MINUTES (or 3 minutes if being radar vectored), then adjust level/speed in accordance with filed flight plan. Proceed to destination radio aid and commence descent at expected approach time (EAT) or ETA.

4. EMERGENCY DESCENT PROCEDURES IN CONTROLLED AIRSPACE

When an aircraft is forced to execute an immediate emergency descent (e.g. rapid cabin depressurization): 1. Turn immediately 30° (or 45°) left or right to clear the airway centerline. 2. Transmit distress message: 'MAYDAY MAYDAY MAYDAY [Callsign] EMERGENCY DESCENT'. 3. Squawk Mode A Code 7700; select emergency frequency 121.5 MHz. 4. Turn on all exterior aircraft lights to maximize conspicuity.

5. DISTRESS AND URGENCY RADIOTELEPHONY COMMUNICATIONS

Category & Spoken Prefix	Operational Definition & Circumstances	Transmission Contents & Radiotelephony Sta
Distress (MAYDAY)	Grave and imminent danger requiring immediate assistance.	MAYDAY MAYDAY MAYDAY, Callsign, Position, Level, Nature of distress, Intentions. Absolute priority.
Urgency (PAN PAN)	Condition concerning safety of aircraft/person, not requiring immediate assistance.	PAN PAN PAN PAN PAN PAN, Callsign, Position, Level, Nature of urgency. Priority over all except distress.
Medical Transport	Aircraft protected under Geneva Conventions transporting medical personnel/supplies.	Prefix 'PAN PAN MEDICAL' announces protected status.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Does transmitting the phrase 'MINIMUM FUEL' give the aircraft priority over other traffic?

- [A]: Yes, immediate clearance to land.
- [B]: No, it is not an emergency and confers no priority.
- [C]: Only if flying in Class A airspace.
- [D]: Yes, priority over all VFR flights.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 'MINIMUM FUEL' indicates that the flight cannot accept undue delays, but confers NO PRIORITY. Priority is only granted by declaring 'MAYDAY MAYDAY MAYDAY FUEL'.

Q2: In IMC, following a communications failure while being radar vectored, how long must the assigned speed be maintained?

- [A]: 3 minutes
- [B]: 7 minutes
- [C]: 15 minutes
- [D]: Until reaching destination

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Under SERA.8035, if under radar vectors, maintain speed/level for 3 minutes (compared to 7 minutes if on a published route).

Q3: Within what distance of the runway threshold must ATC cease applying speed control?

- [A]: 2 NM
- [B]: 4 NM
- [C]: 8 NM
- [D]: 10 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Speed control shall not be applied or continued within 4 NM of the threshold on final approach.

Q4: Which distress call prefix is used when an aircraft is threatened by grave and imminent danger and requires immediate assistance?

- [A]: PAN PAN
- [B]: MAYDAY spoken three times
- [C]: SECURITE
- [D]: EMERGENCY

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The international spoken distress signal is 'MAYDAY' spoken three times.

Q5: If an aircraft suffers complete radio failure in VMC, what action should the PIC take?

- [A]: Squawk 7600 and continue to destination at assigned Flight Level

- [B]: Continue in VMC, land at the nearest suitable aerodrome, and notify ATS expeditiously
- [C]: Circle in present position for 45 minutes
- [D]: Descend immediately to 500 ft AGL

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under SERA.8035, if in VMC, the aircraft must remain in VMC, land at the nearest suitable aerodrome, and report to ATS.

Q6: In radiotelephony phraseology, what standard word indicates that a transmission has ended and no response is expected?

- [A]: ROGER
- [B]: OVER
- [C]: OUT
- [D]: WILCO

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. 'OUT' signifies that the exchange is finished and no answer is expected ('OVER' signifies my transmission has ended and I expect an answer).